

Stress shift and the nucleus

CARLOS GUSSENHOVEN

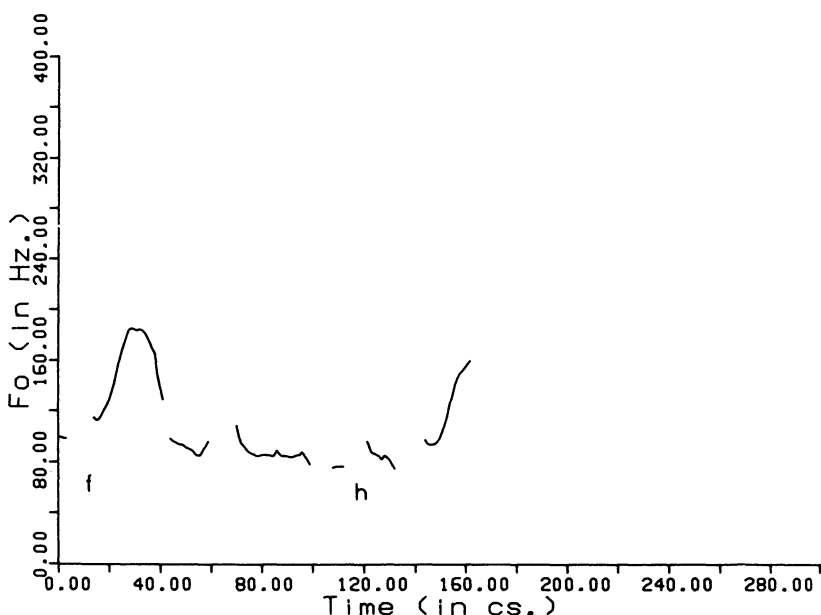
Abstract

This article investigates the phonological relevance of the nucleus of the intonation contour. It does so primarily by reviewing the conditions under which stress shift occurs in English and Dutch, to see if such rules are in any way sensitive to the feature [nuclear]. One Dutch rhythm rule is identified which must refer to the feature. An attempt is made to relate the findings to the distinction claimed to exist in British English between a simple (mononuclear) fall-rise and a compound (binuclear) fall-plus-rise. It will be argued that no such distinction exists in either English or Dutch. It is pointed out that the existence of phonological rules that refer to the final sentence accent of a tone group does not imply that nonfinal sentence accents are distinct from the final one in a semantic sense.

1.0. Introduction

Consider the following British English contours:

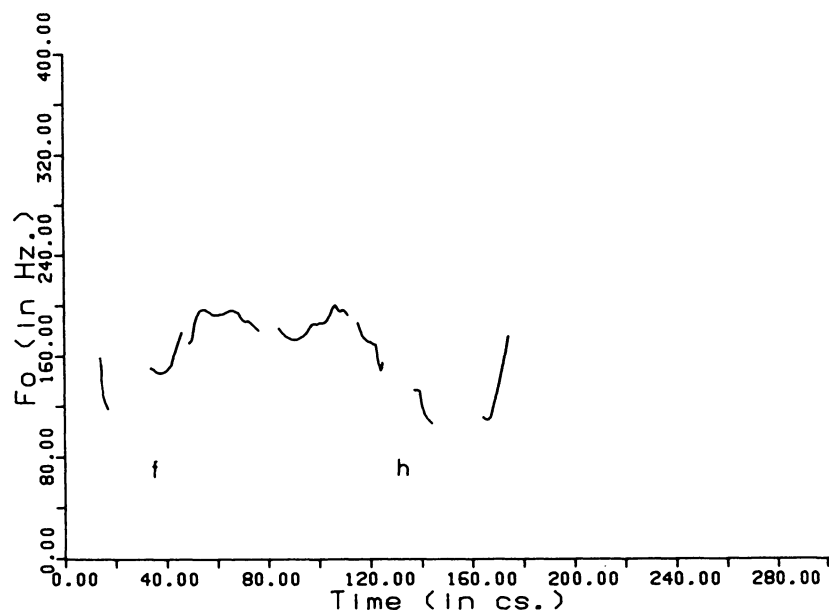
(1) (It's no use calling your father)



Your father won't be able to help me.

(Call your uncle!)

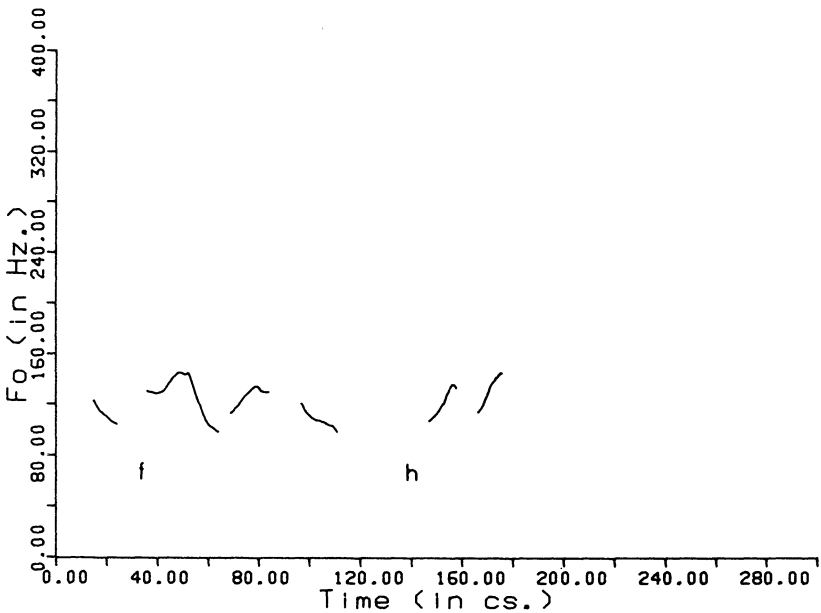
(2) (WHAT's that you said?)



Your father won't be able to help me?

(That I do not believe)

(3) (Forget about it, you silly!)



Your father won't be able to help me.

(Although he may be able to waste a lot of my time....)

In each of (1), (2), and (3) there would seem to be at least two syllables, /fa:/ and /help/, that somehow stand out as the result of pitch features.¹ Also, in each case, the first syllable of *father* seems to be prominent because of a step-up in pitch relative to *Your*, while *help* seems prominent at least partly because of the higher pitch on *me*, in addition to whatever its pitch is relative to *to*. These are what we might call the phonetic facts, and few linguists would dispute them. This, however, is probably the extent to which linguists can be made to agree when it comes to either the semantic function of these prominence peaks, or the phonological structure that underlies their presence. Are the peaks interdependent, in that a speaker must use the first pitch feature in combination with the other, or can he choose the one independently of the other?

Brown et al. (1980), based on their finding that listeners, who could be characterized as non-naïve in one case and as phonetically trained in another, frequently hear two 'tonics' in utterances like these, would argue that the two characteristics that are traditionally attributed to the 'tonic', viz. that of marking new or significant information, and that of falling on the last lexical item in tone-group-like stretches of speech representing

'unmarked' utterances (Halliday 1967), should properly be seen as resulting from two separate systems: one marking the salient information peaks and another marking the final boundary of the tone group.²

By contrast, Liberman and Sag (1974) would claim that the structure of their 'contradiction contour', here reproduced as (4), is a holistic unit. They base this on their observation that listeners are able to interpret the meaning 'contradiction' even when the contour is hummed or played on a musical instrument. That is, when divorced from its lexical underpinning, the essential meaning of the contour is not lost, provided the general context remains intact. Their 'canonical' example is

(4)



Elephantiasis isn't incurable!

(For a critique of this analysis, see Cutler [1977] and Ladd [1980: 15–6]). I found it difficult to elicit this contour from British informants, who tended to come up with (3).

There is a third view, particularly associated with the British tradition of intonation description (e.g. Palmer 1922; Kingdon 1958; Halliday 1970; Crystal 1969; O'Connor and Arnold 1973), which maintains that intonationally prominent syllables are one of two types: 'nuclear' or 'non-nuclear'. Nuclear prominence owes its distinctive status to the fact that it is found on the first syllable of the stretch of speech associated with one of the 'nuclear tones' of an intonational tone system (fall, rise, fall-rise, rise-fall, etc.), while non-nuclear prominence is found on the first syllable of the stretch of speech associated with a contour from a different system, that of 'head contours' (falling, rising, high level, low level, etc.). Any well-formed intonation contour must have a nuclear syllable, or 'nucleus', which MAY be preceded by a non-nuclear syllable marked by the head contour.³ In view of the extensive literature written in this tradition, it is not difficult to analyze the contours (1) to (4) in terms of the British model.

Utterance (1) would be said to have a fall-rise nuclear tone on *father*, which is spread over the stretch extending from *father* (the fall element) to *me*, with the rise element starting after *help*. Postnuclear stretches are known as 'tails'. There is some latitude in how the prominence on *help* should be characterized. Crystal, for instance, distinguishes between

complex and compound tones: a fall-rise as in (1) would not mark a single nucleus, in his terms, but be an instance of a 'binuclear' (compound) tone, with both *father* and *help* being nuclear (1969: 218). Similarly, Halliday might call the movement away from *help* a 'minor tonic', i.e. one always cooccurring with a 'major tonic', while O'Connor and Arnold speak of a 'simple' fall-rise as opposed to a 'compound' fall-rise. It is not fully clear from their description which one (1) should be assigned to (1973: 30). In the most extensive corpus of spoken English analyzed in terms of the British approach (Svartvik and Quirk 1980), tone groups of this sort are normally marked with two nuclear syllables. Note that irrespective of whether analysts include the concept of compound tones in their system, in the case of (1) we are dealing with a single choice from the system of 'nuclear tones', there being no head contour in the utterance.

Utterance (2) would be said to consist of a high head, beginning at *father*, and a (high) rising nuclear tone, beginning at *help*, a combination that O'Connor and Arnold call a 'high bounce' (1973: 44).

Utterance (3) is a fairly common pattern: a falling head — more properly a sliding head because of the pitch-prominent *won't* — starting at *father*, and a fall-rise nuclear tone beginning at *help*. O'Connor and Arnold would consider the contour a 'switchback', like (1) in one interpretation.⁴

Finally, Ladd analyzes the American English (4) as a combination of a falling head (beginning at *elephantiasis*) and a low-rise nuclear tone (beginning at *-cur-*) (1980: 15).

The question that comes to mind, of course, is why the British tradition should prefer this analysis to many others that are possible. Why could one not postulate, say, a 'dipping-head' contour followed by a low-rise nuclear tone in (1), or a fall-plus-rise tone for the pattern in (4) (as Liberman and Sag would like to do)? Phrased concisely, what is the distinction between nuclear prominence and non-nuclear prominence ultimately based on, if not on some ill-defined notion of descriptive convenience?

There are a number of conceivable ways open to anyone who would like to defend the analytic framework. One is to characterize the nuclear tones such that they can be unambiguously identified. In a sense, this is of course what the standard descriptions of British English have done (see in particular Crystal 1969: 221–222). It would appear, however, that in practice that characterization either has not been accepted or has proved unworkable in the context of live data. Another approach would be to give a semantic characterization of the nuclear tones that can be shown to be consistent over different utterances, such that a contour can be analyzed on the basis of its meaning. In view of the difficulty of

establishing intonational meaning (see Cruttenden 1981 for a short survey), this approach seems fraught with difficulties.

There may, however, be a third approach. Clearly, the most important evidence for the phonological reality of hypothesized phonological elements is their participation in phonological rules. To make an analogy, there is no firmer motivation for assigning separate statuses to final fortis and lenis obstruents in English than the fact that the rule governing the duration of the preceding postonset voiced segments (consisting of [+syll] ([+son]) ([+son])) cannot be written without reference to some feature that distinguishes between the two classes of consonant in the structural description of that rule (cf. the durations of /i/ in *niece* and /i/ in *knees*, or of /am/ in *pint* and /am/ in *pined*). Observe, too, that it may be difficult to distinguish final fortis obstruents from final lenis obstruents on the basis of their phonetic characteristics ALONE and that their identification is to a large extent dependent on the nonlocal feature of the duration of the preceding voiced segments. Are there perhaps nonsegmental phonological rules in intonation languages that will have to refer to the feature [nuclear] in their structural description?

In section 2, the rhythm rules of English and Dutch are vetted to see if any of them need to refer to [$\pm$ nuclear]. Before this is done, let us consider what theoretical significance would have to be attributed to finding such a rule. One of the (arguably putative) implications of the British descriptions is that there is only one major pitch-prominent syllable in the sentence, or tone group. There is now abundant evidence, both empirical (Brown et al. 1980) and theoretical (e.g. Gussenhoven 1983a), that this is not the case. Also, 't Hart and Collier's description of Dutch (1975), which is based on the free occurrence of any number of sentence accents, would have to be enriched with patently counter-intuitive notions if it were made to conform to the above implication. Of course, inasmuch as the British descriptions admit the occurrence of 'pretonics' or 'head accents', it could be argued that the inference of 'one major accent per tone group' need not be made. But, as is clear from Bolinger (1958), the notion that prefinal accents should be treated as belonging to a different paradigm is also controversial. In Gussenhoven (1983b), for example, the author shows that it is possible to describe all the head contours in O'Connor and Arnold (1973) as resulting from the linking of sequences of tones that belong to a single paradigm. The only thing we can reasonably be concerned with here, therefore, is to establish whether it is necessary for any phonological rule to 'know' which is the last sentence accent in a tone group. If we can establish this, then the feature [nuclear] will have been shown to be necessary only in this sense.

There may be practical significance, too. If an independent criterion for

establishing which is the last sentence accent in the tone group is found, then it may provide a means of solving controversies which exist within the British tradition, such as whether sentences like (1) contain one or two sentence accents. This question is addressed in section 3.

2.0. Rhythm rules

A potential candidate for an independently motivated rule that is sensitive to the feature [nuclear] may be found in the class of rhythm rules. Such rules shift prominences in utterances around to suit a more nearly alternating pattern. They have been reported for a number of languages (e.g. Liberman and Prince 1977 and Prince 1983 for English; Schultink 1979 for Dutch). As a result of the application of a rhythm rule, the stress on one syllable is shifted onto some other syllable that is further removed from a third stressed syllable. This results in the removal of a nonperipheral prominence peak in some domain and the creation of a peripheral prominence peak in that domain, as in the English phrase *fifteen* 'men → 'fifteen 'men, or in the German noun 'Vor,mittag → 'Vormit,tag. If we call the stress to be shifted the *shiftable* (stress) before the application of the rule and the *shifted* (stress) after it, and call the stress that causes the shift to take place the *fulcrum*,⁵ then what we ideally need is a rhythm rule which applies if and only if the fulcrum is [+nuclear]. It would be desirable, moreover, if that rule was obligatory rather than variable, because if it were, the status of the feature [nuclear] would more clearly be brought out as an 'all-or-none' feature, rather than simply as an additional prominence-lending feature that would increase the likelihood of the rule applying. Thus, in the ideal case, the rhythm rule should be insensitive to the pitch height, the 'emphasis' or general prominence of the fulcrum, the number of syllables that separate the shiftable from the fulcrum, and the speaker's style or speed of utterance. Weaker cases are imaginable, of course. It could, for instance, be the case that the rule is obligatory if the fulcrum is nuclear and variable if it is not. Against this background of desiderata, then, let us look into the conditioning factors of the rhythm rules of two languages, English and Dutch.

2.1. An English rhythm rule

The subject of 'rhythmical variation' or 'stress clash' has long lain dormant in linguistics. This also goes for the English rhythm rule, informally known as the 'thirteen men rule'. It has of course been part of

English language teaching programs at least since the beginning of this century (e.g. Rippmann, 1913 [1906]: 82; Ward, 1945 [1929]: 167) and has been mentioned in the linguistic literature (e.g. Kiparsky 1975: 595), but it was only in 1977 that it received formal treatment in a theoretical framework in the Liberman and Prince article on metrical stress assignment. Until Prince (1983), not much appears to have been done with the descriptive device put forward there to deal with the phenomenon of stress clash: the 'metrical grid'. Thompson (1980) is to be credited with a further specification of just when a clash occurs and has incorporated the device in his 'footmaker', a set of rules generating Abercrombian foot-boundaries; and Giegerich (1983) argues that a more economical solution can be arrived at without invoking the concept of metrical grid at all. By contrast, Prince (1983) claims that the grid can do everything the tree can do and proposes that tree structure be dispensed with. Very briefly, and ignoring problems and implications, Liberman and Prince's metrical grid is an alignment of columns over syllables, whose heights are determined by the metrical tree structure assigned to the utterance on the basis of [$\pm$ stress] markings of the syllables. The feature [stress] is a segmental feature, which would seem to correspond with Gimson's 'strong' and 'weak' vowels (1980: 226). Its values are assigned by SPE-type stress rules (Chomsky and Halle 1968), whose operation is seen to be related to the tree-building procedure, and apply together with this procedure in iterations. The trees are binary-branching and are built up from right to left, assigning s (strong) and w (weak) to the descending branches at each node, according to three principles (cf. Liberman and Prince 1977: 264–268):

1. no s dominates a [-stress] syllable;
2. tree-building proceeds from the right, with maximum ('highest') structures being erected at every [+stress] encountered;
3. the right-hand branch of any node is assigned s if it itself branches (assuming we are dealing with word-level phonology).⁶

On the basis of the tree, a metrical grid is erected via the 'relative prominence projection rule' (RPPR). This grid should be seen as the representation of the hierarchy of relative prominences in the word. If, with Liberman and Prince, we call the strongest terminal node corresponding to each node in the tree its 'designated terminal element' (DTE), then the RPPR requires that for each node the DTE of 'its strong subconstituent is metrically stronger than the DTE of its weak subconstituent'. If strength is symbolized as column height, the metrical grid for *association* is therefore as given in (5). If 'element' is used to refer to any point where columns and levels cross (i.e. the asterisks in [5] ff.), the rhythm rule proposed by Liberman and Prince for English amounts to

(7)

a Chinese expert's advice

The problem with the constraint is that it suggests that the potentially clashing elements are situated in the columns over *-nese* and *ex-*, rather than over *-nese* and some following, stronger syllable, like *-vice* in (7). And of course, if iambic reversal does take place in (7), it is because *-vice* is a fulcrum, not *ex-*. In fact, because they define ‘clashing’ the way they do and constrain the rule the way they do, they predict pressure for change in many situations where there is none, as in (8).

(8)



A Chinese expert is useless.

If the general descriptive framework is to be maintained, the definition of 'clashing' should be revised. This could be done by including a provision that the right-hand column not be lower than the left-hand column, or, alternatively, that the left-hand element be positioned at the top of its column. Prince (1983), whose description does not have the above flaw, in effect introduces the second constraint. In his treeless description it says 'Move the clashing element', and since in (8) this operation would leave a hole in a column, it is ruled out by the general well-formedness condition that columns have elements at all levels.

In summary, then, there is no employment for a feature [nuclear] in the English rhythm rule, because if we say that the shiftable must be [-nuclear], it could be objected that it should simply not be the main stress in the syntactic phrase, irrespective of its position in the intonational phrase. Neither can we say that the fulcrum must be [+nuclear], as it will be objected that [-nuclear] stresses, too, cause stress shift. As there are no other rhythm rules on a level beyond the word in English, we will now turn to the rhythm rules of Dutch.

2.2. Rhythm rules in Dutch

Rhythm rules exist in many forms. While iambic reversal would appear to be the only phenomenon of its kind in English, it would not be reasonable to regard stress shift in Dutch as a unitary process, although, of course, it

may be true that general principles or rhythmic patterning are at the basis of all stress-shift phenomena.

It will be argued that Dutch has four separate rhythm rules that operate on the level of the phrase or beyond.

2.2.1. *Iambic reversal.* The phenomena in Dutch that most closely resemble English iambic reversal should properly be seen as resulting from two separate rules. In general, rhythmic restructuring would seem to be sensitive to the following three factors:

1. Rhythmic template of the word in which the shiftable stress occurs. Thus, *tum-tum* (if followed by an accented word) is less likely to undergo iambic reversal, all else being equal, than *tum-ti-tum*, while *tum-tum-ti* may occupy an intermediate position: *unique* may never undergo it, *fantastic* might undergo it, while *picturesque* is more likely to undergo it than either of these. Similarly, increasing the number of unaccented syllables between shiftable and fulcrum will decrease the likelihood of stress shift, and *tum-tum-ti* will thus be less resistant than *tum-tum-ti-ti*: *bipolar magnet* is more vulnerable than *bilateral talks*. It will be clear that Liberman and Prince's metrical grid and their statement of iambic reversal do not capture these differences in any way (cf. also Bolinger 1981). In addition, it may be the case that the position of the fulcrum in its word may be of influence, as the number of unaccented syllables preceding it in the word may be related to the pressure it can exert on the shiftable. Without controlled observation, however, it is difficult to establish just how important these factors are. Moreover, it will be necessary, for the purposes of a more detailed description, to decide on an operational definition of stress shift in order to deal with cases where the shiftable and shifted positions are sufficiently far removed for both syllables to be perceived as 'stressed', as in the case of '*heterosexual tendencies*'.

2. The presence of a semantically active boundary between the shiftable and shifted positions. Liberman and Prince explicitly mention this factor and propose that in order to make their iambic reversal rule sensitive to such boundaries, a lexical entry should have a guaranteed metrical strength of two levels, even if it is monosyllabic. This requirement blocks iambic reversal in *Montana cowboy*, but it allows it in *good-looking soldier*:

(9)

Montana cowboy

good-looking soldier

3. Familiarity of the concept denoted by the phrase concerned, possibly

measurable as its frequency of occurrence. Thus, when in a talk on brain functions the concept of a *photographic memory* is introduced for the first time, it may well not have iambic reversal applied to it, but subsequent mention will make this increasingly likely. It is suggested that this factor may, on occasion, be exploited by speakers stylistically, to suggest that they 'know what they are talking about'. This may also be the reason why purely evaluative adjectives like *picturesque*, *fantastic*, and *unique* are so much less likely to undergo iambic reversal than purely descriptive adjectives like *Romanesque*, *Norwegian*, and *antique* (cf. also Bolinger's comments on the influence of word meaning, 1981: 40).

While all three factors would seem to have an influence (however they are 'weighted') on the SAME rule in English, in Dutch, factor 2 sharply distinguishes between what I will call *stress retraction* and *rhetorical retraction*.

Stress retraction is virtually obligatory in certain adjectives and adverbs. Schultink (1979) characterizes this group of words as follows:

1. in compounds: *dood'ziek* ('very ill'), *donker'blauw* ('dark blue');
2. in compounded derivatives: *drie'stemmig* ('for three singers'), *onder'gronds* ('underground'; -s is a suffix);
3. adjectives like *aan'staande*, *jongst'leden*, *laatst'leden* (before days of the week: 'next', 'last' [2 ×]);
4. formations with *on-*: *on'juist* ('incorrect'), *onbe'leefd* ('impolite').

If the items under type 3 are listed as exceptions, the generalization seems to be that shiftable and shifted positions should be separated by a (productive) morpheme boundary. In fact, van den Berg (1970) noted that in words beginning with *on-*, retraction seemed to be sensitive to whether *on-* could be replaced with the free morpheme *niet* 'not'. If it is true that the nature of the boundary provides the explanation, then how should it be defined? Items that fail to undergo stress retraction but do have a boundary of some sort form a more varied group than may appear at first sight. Schultink observes that derived compounds (e.g. *eer'zuchtig* < *'eierzucht*, lit. 'crave for honor', i.e. 'ambition'; or (my example) *grond'wet-telijk* < *'grondwet*, lit. 'ground law', i.e. 'constitution') are exempt from the rule. One could add most adjectives consisting of Noun + Verb + present participle suffix, like *zorg'wekkend* 'worrying', *tijd'rovend* 'time-consuming', *angstaan'jagend* 'fear-inducing' (but not *Frans'sprekend* 'French-speaking'), and almost all words that like *eer'zuchtig* have the pattern ...'...ig (whether derived compounds or not), such as *licht'vaardig* 'rash', *recht'lijnig* 'with a single-track mind', *zee'waardig* 'sea-worthy', *zwart'gallig* 'very pessimistic', etc. Also formations with *vriendelijk* are exempt, like *gebruikers'-vriendelijk* 'user-friendly', and items like *ge-lijk'vloers* 'on the same floor', *Franco'foon* 'Francophone'. Thus, there are

such contrasts as *een'baby-zacht* 'huidje' 'a baby-soft skin' and *een baby-vriendelijke* 'luier' 'a baby-friendly nappy'. Intuitively, it would seem as if adjectives that are subject to the rule contain a higher-order boundary (say #) than those that are not (which may have, say, +). In order to see that this intuition is not based on the facts of stress retraction themselves, observe that shifting adjectives cannot serve as constituents of compounds, but that nonshifting ones can. Thus, while a formation like **ultra-baby-zacht* is not only nonexistent, but also ill-formed, the formation *hyper-gebruikers-vriendelijk* is clumsy, but well-formed. Similarly, **on-bovengronds*, **pseudo-doodziek*, **hyper-keihard*, **ultra-Frans-sprekend* are bad, but *ongelijkvloers*, *pseudo-zeewaardig*, *hyper-eerzuchtig*, and *ultra-Francofoon* (e.g. '~Walloons') are fine. It is apparently this # boundary, of which a word may contain only one, that stress retraction is sensitive to.

Rhetorical retraction applies to adjectives and adverbs irrespective of a boundary (provided of course the usual phonological conditions are met). Both the absence of any discussion in the literature of this rule and the occasional complaints against its effects in the letters-to-the-editor sections of general periodicals suggest that the rule is an innovation. As its name implies, style, or more generally the rhetorical or propagandist content of the text, would appear to be an important variable accounting for its application. Its effects can frequently be observed in the speech of politicians, participants in meetings, and radio and television announcers, as in examples (10), (11), and (12) respectively.

- | | | |
|------|-------------|--|
| (10) | poli'tiek | -een 'politieke 'zaak ('a political issue') |
| | di'rect | -'directe 'steunmaatregelen ('direct cash aid') |
| (11) | for'meel | -'formeel over'leg ('formal negotiations') |
| | princi'peel | -een 'princiële be'slissing ('a decision in principle') |
| (12) | vo'kaal | -het 'vokaal en'semble ('the vocal group') |
| | popu'lair | -de serie 'populaire klas'sieken ('the popular classics series') |

It should therefore be possible to find a minimal pair of adjectives that differ only in the presence of a morpheme boundary. An example is *onver'mijdelijk* in the sense of 'inevitable; always there' (no boundary) and *onver'mijdelijk* in the sense of 'unavoidable; which could not be avoided' (with boundary). In (13) *onvermijdelijke* CAN only mean 'always there', while in (14) we could have either 'the error (*four*) that could not be avoided', with stress retraction, or the same meaning as in (13), with rhetorical retraction.⁷

- (13) de onver'mijdelijke 'fout

(14) de 'onvermijdelijke 'fout

While it may be established that a description of stress retraction is certainly not complete when we have only Liberman and Prince's metrical grid and iambic reversal rule at our disposal, and that much more careful research is needed for both English and Dutch to establish just what the relevant favoring factors are and how they are weighted, we cannot detect any contribution on the part of the feature [nuclear] in the structural description of either stress retraction or rhetorical retraction in Dutch. Our attention should therefore be directed to a third rhythm rule in this language.

2.2.2. *Trochaic reversal.* The rhythm rules discussed above involve stress shifts to the left. Liberman and Prince cite some of Kiparsky's examples of the reverse phenomenon in German (Kiparsky 1966), in which language we not only get *halb'tot* versus *der 'halbtote* 'Man, but also 'anziehen versus *den 'Rock an,ziehen*. This reverse stress shift, which I will refer to as *trochaic reversal*, also exists in Dutch. Kiparsky gives two situations in which it may occur. First, it may occur in compounds (adjectival or nominal), as in 'sichtbar, but 'unsicht,bar (compare 'logisch and 'un,logisch) and in 'Althoch,deutsch, 'Nachmit,tag, 'Urgross,vater, 'Vorur,teil, 'Luftab,wehr, etc. In these situations, a word (simple or compound) with the primary accent on the first syllable is embedded in a larger structure requiring the primary accent on the syllable preceding it, which syllable acts as a fulcrum shifting the embedded primary, now secondary, stress onto a later syllable, which, in Liberman and Prince's terms, must of course be [+stress]. Phonetically, trochaic reversal amounts to a durational restructuring of the postaccentual syllables and does not involve shifts of pitch peaks, as do the iambic rules discussed elsewhere in this article. In Dutch, this trochaic reversal is common in nominal compounds, provided the embedded item is not itself compound and the shifted syllable contains a short (or [-tense]) vowel. Examples are 'tandpas,ta ('toothpaste'), 'bloedli,chaampjes ('corpuscles'), 'jeugdher,berg ('youth hostel'), plant'soenaar,beider ('municipal gardener'), 'kleurpot,lod ('colored pencil'), but not 'dans,leraar ('dancing master'), 'reis,schema ('itinerary'), 'post,giro ('Postal Giro'), which have a long (or [+tense]) vowel in shiftable position.

Second, it occurs in prefixed verbs that have the accent on the (separable) prefix, as in German 'mitnehmen versus *das 'Buch mit,nehmen*. Similarly, Dutch has 'opbreken versus *de 'weg op,breken* ('to break up the road') or 'opmaken versus *de 'kas op,maken* ('to count/make up the cash'). The restriction to verbal prefixes would certainly seem sensible for Dutch,

because 'opmaken in the sense of 'to finish, to deplete', where *op* is more properly an adverb, would not as readily be so treated. Thus, although 'opmaken in isolation is ambiguous, it is not in postprimary position: 'to (completely) spend the cash' would be *de 'kas ,opmaken*. Note that this generalization not only captures the fact that 'opmaken ('to count, make up') is a single lexical entry, but also that trochaic reversal does not operate across a boundary within a nominal compound. As was said above, if the embedded element is itself a compound, no stress shift takes place: '*spek ,pannekoek* ('bacon pancake'), '*padvinders ,zakmes* ('scout penknife'), '*oven ,handschoen* ('oven mitt').

The question we need to consider here is whether trochaic reversal is in any way sensitive to the feature [nuclear]. Examples (15) and (16) demonstrate that it is not. In (15) the nuclear syllable would be *nooit* ('never'), while in (16) it is *geld* ('money').

- (15) ~~Geld oppotten doe ik nooit.~~
money hoard do I never
- (16) ~~Nooit zal ik mijn geld oppotten.~~
never will I my money hoard

In either case, the prominence on *geld* is the fulcrum, and in either case *-pot-* is the most prominent syllable of *oppotten* (in isolation '*oppotten*'). That is, given the right lexical input, trochaic reversal is operative irrespective of the value of the feature [nuclear] for *geld*. It is of course true that *op* may not be nuclear, but as in the case of iambic reversal, this does not compel us to include it in the structural description or add it as a constraint. We are again dealing with an if-condition, not an iff-condition: the requirement is simply that the fulcrum be metrically stronger than the shiftable.

2.2.3. *Non-nuclear retraction.* We have now come to the last of the four Dutch rhythm rules. Compare (17), in which the adjective compound *licht-blauw* ('light blue') occurs in sentence-initial position, with (18), in which the adverb *meestal* ('as a rule') occurs in the same position. Note that both *licht-blauw* and *meestal* have the word accent on the final syllable in isolation.

- (17) a. Licht-'blauw is de kleur van haar 'ogen.
b. *'Licht-blauw is de kleur van haar 'ogen.
light blue is the color of her eyes
- (18) a. *Mees'tal gaan ballonnen naar 'boven.
b. 'Meestal gaan ballonnen naar 'boven.
as-a-rule go balloons (to) upwards

Utterances (17) and (18) have the same rhythmical pattern, except for the first two syllables: stress shift does not apply to *licht-blauw*, but does apply to *meestal*. The reason why *licht-blauw* does not undergo it is that for the purposes of stress retraction there is no fulcrum nearby, and certainly not in the same phrase. The stress shift for *meestal* is obligatory: a pronunciation with the word accent on the second syllable would be deviant, given the intonation contour. Clearly, we are dealing with a different rule altogether. Its distinguishing characteristics can be stated as follows:

1. While rhetorical retraction, and to a lesser extent stress retraction, are variable, this rule is obligatory. Both nonapplication in the right context and application in the wrong context are felt to be deviant by native speakers.

2. Give or take a word boundary, the stress-shift rules we have seen so far apply to certain rhythmic configurations or metrical grids, rather than to particular morphemes or words. The present rule applies to a finite set of quantifiers and adverbs, provided their phonemic makeup includes at least two [+stress] vowels and the word accent falls on the last of them.

3. In the description of the stress-shift rules discussed so far we needed to refer to constituent boundaries and the syntactic bond or distance between fulcrum and shiftable. The present rule is applicable with the fulcrum following AT WHATEVER REMOVE and with WHATEVER BOUNDARIES occurring between it and the shiftable, provided it follows in the same tone group; this fulcrum is the nucleus.

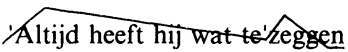
The following items are subject to the rule, which we will refer to as *non-nuclear retraction*:

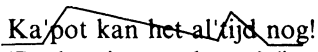
(19) Quantifiers	Adverbs
<i>allebei</i> ('both')	<i>altijd</i> ('always')
<i>allemaal</i> ('all')	<i>allesbehalve</i> ('anything but')
<i>allerlei</i> ('all sorts')	<i>enigszins</i> ('somewhat')
<i>iedereen</i> ('everybody')	<i>helemaal</i> ('entirely')
<i>menigee</i> ('many people')	<i>meestal</i> ('as a rule')
	<i>niettemin</i> ('nevertheless')
	<i>ruimschoots</i> ('amply')
	<i>overal</i> ('everywhere')
	<i>onlangs</i> ('the other day')
	<i>(tot) dusver</i> ('thus far')

The input to the rule is lexically determined. It is not the case that all words in these classes are subject to it. There are a few that have the accent on the last, but do not shift, like *alleen* ('only, alone'), *althans* ('at least'), *inderdaad* ('in fact'), *uberhaupt* ('at all'). Interestingly, however, the words *bijna* ('almost') and *onderling* ('among one another') have a

'dictionary' pronunciation with the accent on the first syllable, but are frequently pronounced with the accent on the last when they occur as one-word sentences (and as such carry the nucleus). If these 'linguistic' pronunciations come to replace the 'dictionary' pronunciations, which does not seem unlikely, we would have two instances of accentual back-formation. The active status of the rule is also illustrated by the fact that beginning Dutch learners of English often pronounce words like *always*, *also*, *beforehand*, *nowadays* with the accent on the last syllable when they occur in nuclear position. Finally, it may be noted that *al* ('already') which derives from earlier *alreeds*, *alrede*, CANNOT be given the nucleus, the reason for which is possibly that it has lost the syllable that is to carry it. The equivalent of English *I've done that alREADY* is *Dat HEB ik al gedaan*, with the accent on the finite verb form. It cannot occur as a one-word sentence either, it being obligatorily accompanied by *nu* ('now') or *toen* ('then') to carry the nucleus. Similarly, *wel eens* ('occasionally') is excluded from nuclear position, *eens* being a fossilized 'weak form' with reduced schwa, although the pressure of non-nuclear retraction frequently brings some speakers to put the nucleus on that syllable when *wel eens* occurs in a one-word sentence.⁸

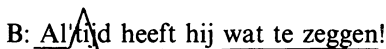
It should be stressed that the rule is sensitive to the feature [nuclear] ONLY. The degree of emphasis that is applied to *altijd* in (20) is entirely irrelevant: provided the nuclear tone follows (on *zeggen*), the word accent comes on *al*-. Similarly, no amount of muffling of the prominence on the word *altijd* in (21) will cause it to have the accent on *al*-, provided the word has the nuclear tone, in this case a fall. That is, both (20) and (21) have a high falling 'head accent' on the first prominence peak, and a low fall on the second:

(20)  *Altijd heeft hij wat te zeggen*
'He's always got to say something.'

(21)  *Ka'pot kan het al'tijd nog!*
'Broken it can always' (i.e. stop fiddling about with it).

Neither is the rule sensitive to whether the word comes early or late in the tone group. When we have an early nucleus in (22), for instance, the word accent is on the second syllable:

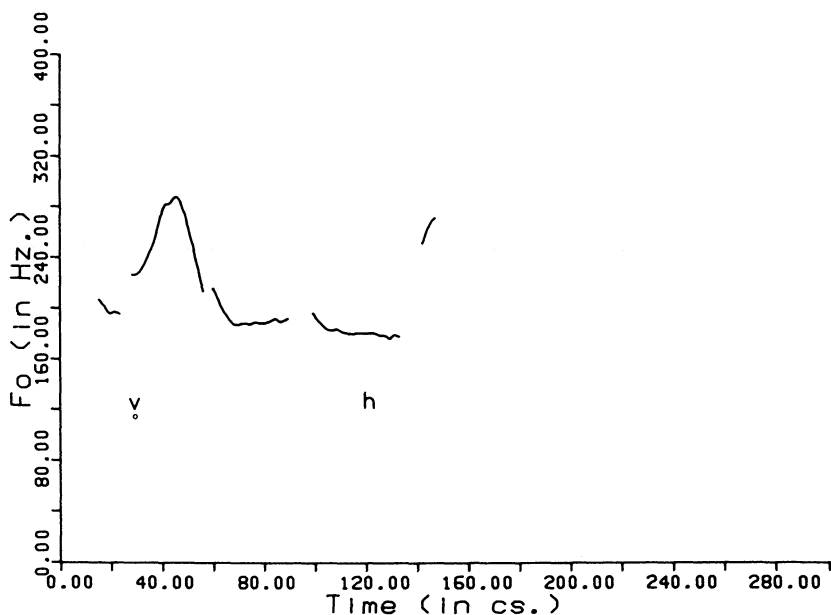
(22) (A: *When* has he got something to say?)

B:  *Altijd heeft hij wat te zeggen!*

What we are dealing with is, in other words, the strongest possible case of a rhythm rule sensitive to the feature [nuclear]: (a) it is obligatory, and (b)

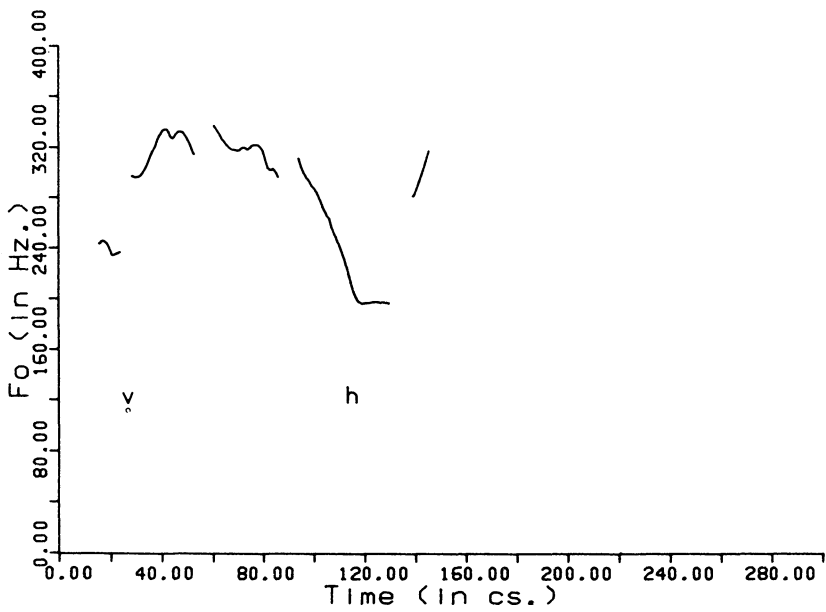
it applies if the fulcrum is nuclear (and, by necessity, the shiftable is not). Obviously, it would be interesting to see what this rule can tell us about the intonational structure of the utterances that we began this article with. Utterances (1), (2), and (3) in fact translate quite naturally into Dutch, without their contours being affected in any important way. (Examples [23]–[28] were elicited from a Dutch female informant who was asked to pronounce the Dutch sentences concerned with the intonation patterns of the English utterances (1), (2), and (3). Note that Dutch /h/ is voiced.)

(23)



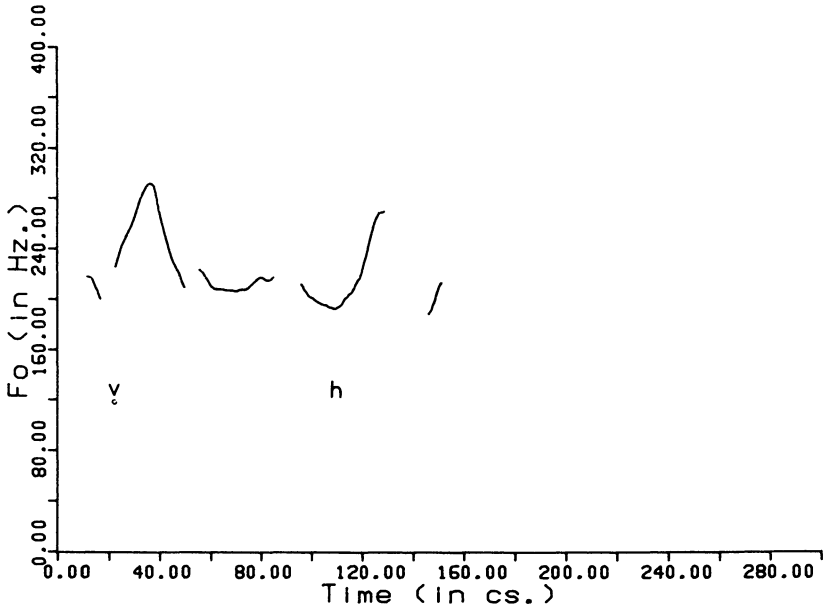
Je vader zal me niet kunnen helpen.

(24)



Je vader zal me niet kunnen helpen.

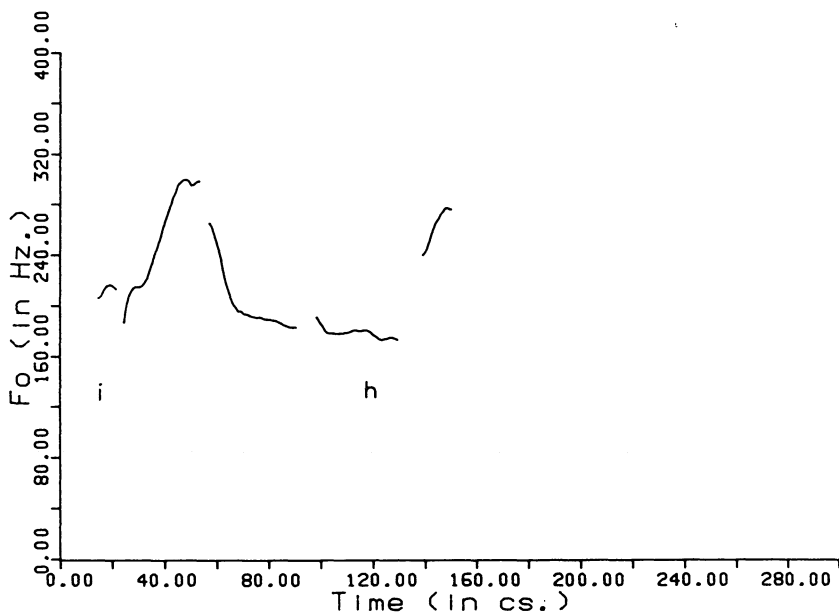
(25)



Je vader zal me niet kunnen helpen.

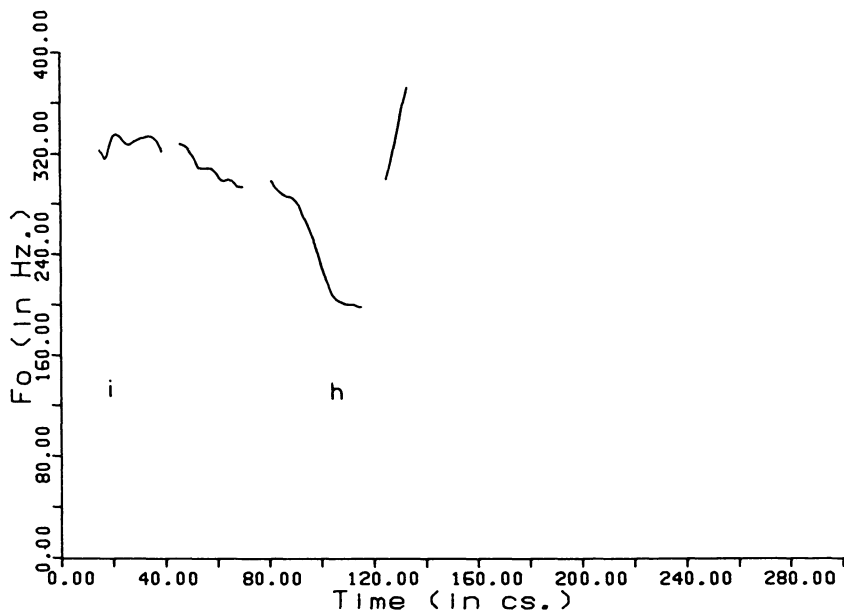
When we substitute *iedereen* (subject to non-nuclear retraction) for *je vader*, we get:

(26)

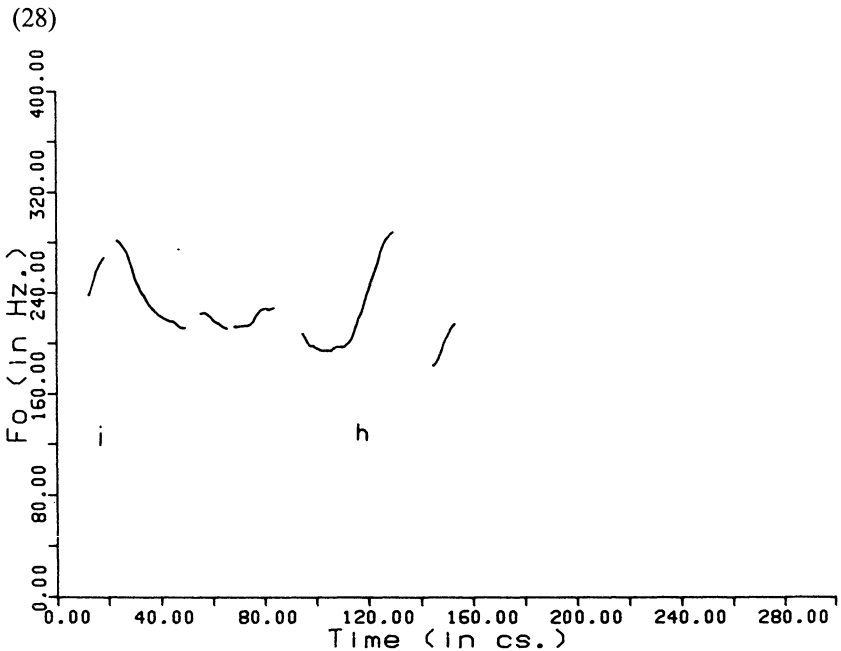


Iedereen zal me niet kunnen helpen.

(27)



Iedereen zal me niet kunnen helpen.



Iedereen zal me niet kunnen helpen.

When we compare (26) with (27) and (28), it will be clear that stress shift has occurred in (27) and (28), but not in (26). Clearly, *helpen* is postnuclear in (26) — and by consequence in (23) — and the rising pitch movement does not mark a sentence accent. In (24), (25), (27), and (28), however, *helpen* is marked by an accent-lending rise, as is evident from the fact that non-nuclear retraction has taken place in (27) and (28).

Unfortunately, non-nuclear retraction is a rule of Dutch, not of English, and the intonation of Dutch is not the same as the intonation of English. Willems (1981) demonstrates that both English and Dutch listeners can fairly successfully tell a Dutch contour from a corresponding English one, and Gussenhoven (1983a) gives illustrations of how the languages may differ even with regard to the location of the nucleus. Yet I would like to suggest that at the level we are dealing with here, that of the phonological status of major accents in the tone group, the two languages do not differ, and that it is therefore not unreasonable to see non-nuclear retraction as reflecting the phonological relevance of the nucleus in a shared system. There are, to be sure, other nucleus-governed phenomena that the languages have in common. One, pointed out to me by Bob Ladd (personal communication), is downstep, a term borrowed from analyses of tone languages, to describe what O'Connor and Arnold (1973: 37) call

the 'stepping head': a series of descending terraces, the beginning of each one of which marks an accented syllable (Pierrehumbert 1980). 't Hart and Collier describe the same phenomenon in Dutch as a sequence of E falls (1975). Observe that while the domain for downstep can be — and often is — smaller than that of the tone group, it cannot be terminated in the middle of an NP or VP: its termination must coincide with a major category boundary. Thus, in (29), downstep cannot end after *damn*:

(29) *The whole damn business will fall THROUGH.

However, the nucleus always provides a possible point at which downstep may terminate. Although *fall through* is a single lexical item, *through* may fall outside the domain of downstep in (30) because it is the nucleus:

(30) The whole damn business will fall THROUGH.

When we add another word to carry the nucleus, it can no longer terminate downstep in the same way, as is illustrated by

(31) *The whole damn business will fall through TRAgically.

The same regularity can be observed in Dutch: while it is possible to leave *toestanden* out of the downstep domain in *De hele familie van der Meer had van die moeilijke TOEstanden* ('The whole van der Meer family had these difficult problems'), downstep cannot be broken off at the same point in *De hele familie van der Meer had van die moeilijke toestanden geNOEG* ('... was fed up with these difficult problems').

3.0. The divided fall-rise: one or two nuclei?

The question we can now try to address is whether Dutch non-nuclear retraction can tell us anything about postnuclear prominence, more particularly whether a case can be made in Dutch for the existence of a simple fall-rise by the side of a compound one, which is the 'simple vs. compound' opposition most frequently postulated for English (the only one, for instance, to be included in O'Connor and Arnold's description). To take O'Connor and Arnold's examples as the basis for our discussion, let us see whether the putative distinction in (32) and (33) (1973: 29) can be shown to have a counterpart in Dutch:

(32) My[˘]mother was born in Sheffield (but not my father).

(33) My[˘]mother was born in Sheffield (isn't that interesting?).

They point out that although the resultant contours are 'very similar indeed', 'they are nevertheless very different in meaning' and that it is

therefore necessary to keep them separate. Again, both utterances translate fairly literally into Dutch and retain their intuitively different meanings. And, barring low-level phonetic detail with respect to pitch span (Willems 1981), their contours also remain intact, with the fall and the rise taking off from corresponding elements. In the notation of 't Hart and Collier (1975), the pattern for both would be⁹

- (34) *Mijn moeder komt uit Sheffield.*

0 1&A 0 0 0 0 2

(0 = level, 1 = prominence lending rise, A = prominence lending fall, 2 = nonprominent rise.)

If we now move one of the words that is subject to non-nuclear retraction into the position of *Sheffield*, by replacing the lexico-syntactic structure with, say, *Mijn moeder kent ze allemaal* ('My mother knows all of them'), then the version corresponding to (32), in which only *mother* is claimed to be nuclear, ought to show the rise taking off from the first syllable of *allemaal* if it, too, is [-nuclear] in Dutch, whereas in the equivalent of (33), in which both *mother* and *Sheffield* are claimed to be nuclear, we ought to see a rise taking place on the last syllable of *allemaal*, if indeed it is nuclear. Both versions, however, are identical.

- (35) *Mijn moeder kent ze allemaal*

0 1&A 0 0 0 0 2

(either 'but not my father' or 'isn't that interesting?')

and have the rise on the final syllable.

Moreover, this final rise is not a nuclear tone, but a boundary tone, as is apparent when we add a further unaccented word like *nog* ('still') or *natuurlijk* ('of course') to this sentence. Such a contour is distinct from one which has a nuclear rise on *allemaal* ([36]; cf. Liberman and Sag's 'contradiction contour'):

- (36) *Mijn moeder kent ze allemaal nog!?*

0 1&A 0 0 0 0 0 1 | θ (or 2)

(θ = high reference line)

- (37) *Mijn moeder kent ze allemaal nog.*

0 1&A 0 0 0 0 0 2

For Dutch, in other words, it would appear that no case can be made for a distinction between a simple and a compound fall-rise. The fall element is treated as nuclear and the rise element as non-nuclear in either case.

It should again be stressed that this conclusion concerns a fact of Dutch, not of English. In spite of the close resemblance of the intona-

tional systems of the two languages, one cannot use the absence of an option in one language as evidence that its claimed presence in another is spurious. Yet, the fact that the English pair (32)–(33) has phonetically very similar — and semantically equivalent — counterparts in Dutch, which appear to be identical, should make one suspicious. The problem is, too, that in at least some discussions about the opposition between the fall-rise and the fall-plus-rise in English, the case for it is artificially strengthened by the tacit inclusion of contours that consist of a prenuclear accent (a falling head) followed by a nuclear rise (cf. also note 10). Sharp (1958), who offers a large number of examples of the opposition, would on occasion seem to smuggle in such examples with nuclear rises. Examples of ‘true’ oppositions he gives are

- (38) ˘He’s the plumber (= the other man isn’t).
 ˘He’s the plumber (= ask *him* to mend the tap).
 (39) ˘I can (= even though *you* can’t).
 ˘I can (as an unenthusiastic reply to ‘Who plays bridge?’)

But the following example should be treated with caution:

- (40) ˘What shall I tell him? (= I really can’t think of anything),

which is probably ‘*What shall I TELL him?*’ This is certainly true for ‘*All right*’, which is — normally speaking — impossible with the nucleus on *all*. Observe that there is, of course, a real distinction between (not Sharp’s examples):

- (41) ˘TWELVE men (= not eleven men).
 (42) ˘Twelve.MEN (= don’t you mean eleven women?),

as can readily be seen if we replace *twelve* with *thirteen*:

- (43) Thir˘TEEN men (= [41]).
 (44) ˘Thirteen .MEN (= [42]).

But how are we to split up (43) into two phonologically different patterns? Of course, it is possible to give somewhat different CONTEXTS in which (43) would be appropriate. One context is given in (41). Another would be created if we assume it to be spoken in a surprised or tut-tutting fashion by someone who receives a photograph of her friend’s wedding ceremony, in which she appears before a clerical official with thirteen bridegrooms. What would appear to constitute the difference here is the pragmatic basis for the choice of a nonfocus status for *men*: in the first example, *men* was already present in the discourse, in the second it was not. Rather, it was available for nonfocus reference in view of the background of the wedding, as her friend was expected to marry a man. It does not follow,

however, that the two structures are also phonologically distinct.

There is a further problem that the Dutch contour in (34) raises. Standard descriptions of British intonation have it that the rise element of the fall-rise takes off from the last 'secondary accent' in the tail, rather than on the last syllable of the tone group. Thus, Gimson (1980: 268) says it is 'initiated on the syllable carrying the secondary accent', and O'Connor and Arnold (1973: 13) state that it takes place 'on or from the last stressed syllable of the word group'. Crystal (1969: 220) speaks of 'distribution (of the elements) over lexical items', implying association of the rise element with words rather than syllables. (Incidentally, Crystal's distinction between 'complex' and 'compound' fall-rises is easier to interpret: the former are confined to the syllable [1969: 217], the latter are spread over more words, and both [32] and [33] exemplify compound tones in his terms. Crystal's concern is to separate compound tones from sequences of tone groups.) Thus, the situation in British English would seem to be different from that found in Dutch, but also from that found in American English. Pike (1945) associates the rise element ('-3') with the last syllable of the tone group, irrespective of word accent:

- (45) Help, Catherine!
 1 -4/-4- -3/

Similarly, Liberman (1975) and Pierrehumbert (1980) explicitly state that their high boundary tone, one function of which is to provide the rise element of the fall-rise, can only be associated with boundaries of the intonation contour.

In order to demonstrate that we are dealing with a real difference between British English on the one hand and American English and Dutch on the other in the manner in which the final rise is associated, we need examples in which the 'secondary accent' is at least two syllables removed from the end of the tail, since the fact that the word accentual patterns of, for instance, *Glasgow* and *Dundee* are preserved in (46) and (47) could be due to the preservation of the durational structure, accented syllables being longer than unaccented ones, all else being equal. That is, we may hear the 'accent' on *dee* in (47) as a result of the greater durational latitude this syllable gives to the final rise, as opposed to the shorter *gow*, which may cause the rise to take the shape of a step-up from *Glas-* in (46).

- (46) My[˘]mother was born in Glasgow.
 (47) My[˘]mother was born in Dundee.

In an utterance like (48), however, where the secondarily accented syllable *cow-* is more than two syllables removed from the end, we would have, crudely, pattern (a) according to the standard British description, but

pattern (b) according to an alternative description which associates the second element of a fall-rise with the last syllable:

(48) a. ~~That's what I call cowardliness!~~

b. ~~That's what I call cowardliness!~~

Informal observation suggests that informants accept both patterns as well formed and find it difficult to hear any difference between them. In four cases out of five, moreover, informants confidently responded with pattern (b) when offered (a) for imitation. It is suggested, therefore, that the difference is of a low-level phonetic kind, and that structurally the two utterances in (48) are identical as far as nuclear tone choice and location is concerned. Certainly, I have found no claims that this difference should be associated with a difference between simple and compound fall-rises.¹⁰

There is, moreover, language-internal evidence in English for a unitary analysis of a (divided) fall-rise that is at least suggestive. Echoed statements, in the words of O'Connor and Arnold (1973: 75) may have the 'effect of questions in most cases, as in so many other European languages [and are] used ... to elicit a repetition by the listener of something he has said'. This type of utterance requires a high rise as the nuclear tone. Note that for (1) that nuclear tone would have to go to *father* rather than *help*, and for both (32) and (34) to *mother* rather than to *Sheffield*, as in (49) and (50). Similarly, an impatient repetition, an utterance type not dealt with by O'Connor and Arnold, requires a (high) fall on the nuclear syllable. Note again, that if (1), (32), and (33) were repeated by a speaker who wished to register his impatience at not having been understood the first time, this fall would likewise be placed on *father* and *mother* respectively, as in (51) and (52):

(49) Your *father* won't be able to help you?

(50) Your *mother* was born in Sheffield?

(51) Your *father* won't be able to help you!

(52) My *mother* was born in Sheffield!

It could of course be argued that in the case of double nuclei the first element rather than the second element is selected for these noncomplex tones because, as both Halliday and O'Connor and Arnold claim, it is the more important of the two. This approach leads to obvious problems, however. Note that anyone who claims that the tones in (32) and (33) are different on the basis of the semantic difference between the utterances would also have to argue that their impatient counterparts or their echoed counterparts really contain different tones, as the semantic differences between them remain unaffected. Again, it would appear that we are here

dealing with the straightforward substitution of one term for another from the same paradigm. The tests also suggest that part of Crystal's discussion of the compound fall-plus-rise may be based on tone groups consisting of a falling head followed by a rising nuclear tone, as in /RÒUND the CÒRner/ (1969: 268), although, of course, given out of context as they are, it is not really possible to pronounce on their structure. In any event, this would provide an explanation for his statement that the second element of a fall-plus-rise is functionally more important than the first (1969: 219).

There does, however, remain a problem even for an analyst who wishes to see a divided fall-rise as one single accent, in particular when there is a further hypothesis lurking round the corner: one that would claim that tones have meaning. Somewhere a boundary must be drawn between a sequence of two tone groups, one with a fall and one with a rise, and a single tone group with a (divided) fall-rise. What, for example, is that structure of (53) to (60)?

- (53) I'd `do it if I were ,you.
- (54) You're not a Jehovah's `witness, ,are you?
- (55) I can give you a `rise if that's what you ,mean.
- (56) That's what I've always `wanted ... a ,child.
- (57) They never do what you `want them to do, these ,children.
- (58) I said `no', ,damn it.
- (59) `No', he ,said.
- (60) He `failed, the ,numskull.

How can we tell whether we are dealing with a single tone group or a sequence of two? The question, of course, is of interest also when we do not harbor a conviction that tones are semantic units (cf. Crystal, 1975: 29, who points out that they are not necessarily invested with independent meaning, but considers the question phonologically relevant). But if they are, there is the more reason to believe that the answer to the question is nonarbitrary and absolute, and not just one of descriptive elegance. It would, under that assumption, be unexpected for the opposition to be 'neutralized' in utterances like (53) to (60), unless we were to assume that the combination of the meanings of two terms (the fall and the rise) will somehow equal the meaning of a third (the fall-rise). There is perhaps an indicative test that can be applied to these utterances. If we pronounce them with a 'dip' just before the putative 'rise',

- (61) I'd do it if I were you /
- (62) You're not a Jehovah's witness, are you /

it will appear that such dipping is odd in (53), (55), (57), and (59), but is perfectly acceptable in (54), (56), (58) and (60). By this test, the former four would have to be classified as single tone groups with a fall-rise in nuclear position, while the latter four would be sequences of two tone groups, one with a fall and another with a rise.

To return briefly to the O'Connor and Arnold examples we gave at the beginning of this section, it may be observed that although (32) and (33) are identical from the point of view of nuclear tone choice and location, they need not be identical as far as their focus distribution is concerned. It has been observed in many places (e.g. Schmerling 1976: 82; Chafe 1970: 225; Pierrehumbert 1980: 73) that verbs often remain unaccented even though they are 'comment' or 'focus', i.e. unpredictable from the context. Arguably, (32) has just *my mother* in focus, but (33) *my mother was born*. This difference in focus need not be reflected in a difference in nucleus assignment.

It should be observed that an interpretation of the fall-rise as a unitary choice from a paradigm of tones does not make it impossible to regard the rise element as a boundary tone (cf. Brown et al 1980: 117; Pierrehumbert 1980: 16), provided the following points are made:

1. The boundary tone is not an independent choice from a system comprising a boundary fall (or 'low tone') and a boundary rise, but an obligatory movement determined by the nuclear tone option employed for the preceding nucleus;
2. It is not necessarily associated with the last lexical item or any 'secondary stress' in the tail, but — frequently — with the last syllable of the tail.

The double-focus hypothesis concerning the fall-rise has vitiated a lot of research in English intonation. An interpretation of the rise-element of the fall-rise as 'nuclear' and therefore 'focus', forces analysts to look for semantic or pragmatic reasons why the item it is found on is singled out to carry it. A quotation from André (1973–1974: 115) will illustrate the point:

you/see the !difference :N ís that #

As regards the second focus item [i.e. *is*], it can be interpreted as a normally non-focal item, being marked when major focal: in such a case 'be', if the only element in the verb phrase, has contrastive implications in terms of tense and polarity.

The reason for the rise on *is* lies of course in its being final in the tone group concerned. Likewise, Lindström (1978: 102), taking a similar view of the divided fall-rise, is forced to conclude that vocatives may or may not be nuclear (cf. 'Thank you Mr. Smith vs. 'Thank you Mr.. Smith),

thereby missing the generalization that vocatives are appended to tone groups as tails (e.g. Crystal 1975: 25). The same comment applies to Tregidgo (1981), who speaks of 'focus' (for the fall element) and 'counter-focus' (for the rise element) to refer to nucleus + tail combinations marked with a fall-rise, and to Fox (1982) who analyses *they're all the same, these politicians* as having major focus on the part before the comma when *same* has a fall, but as having major focus followed by a minor focus when *same* has a fall-rise, again missing the generalization that postposed nonfocus elements like *these politicians* are appended as tails. (Note the difference between this type of utterance and [57] on the one hand and [56] on the other.) In 1914, Coleman, at the conclusion of an article written well ahead of its time, asked if any of his readers could tell him why it is that *yesterday* is given prominence in (63), but not in (64) (Coleman used scalar numbers, from 1 to indicate low pitch, to 9 indicating very high pitch):

- (63) 1 3 3 4-1 1 1 3 3
 I went to Malvern yesterday.
 1 3 3 3 4-1 1 1 1
- (64) So you went to Malvern yesterday.

Then and now, Coleman's query should receive this answer: the difference in perceived prominence is the result of a difference in choice of 'tone' associated with *Malvern*: in (63) a fall-rise is used and in (64) a fall. The real question, therefore, is why in the one utterance one tone is used and in the other another, the difference in prominence on *yesterday* being incidental to that choice, and not independently of significance.

4.0. Finding the nucleus

The approach taken in this article to establish the identity and the relevance of the nucleus of the intonation contour differs significantly from earlier attempts to give the concept of the nucleus a more clearly defined status by experimental means. In a series of experiments, Currie (1980, 1981) found that listeners identify tonics in isolated utterances on the basis of a number of criteria, among which are the phonetic cues maximum F₀ height, maximum F₀ movement, and maximum intensity. Cumulation of these cues generally led to fairly uniform identification of the tonic by (non-naïve) listeners, but when these cues were in different locations in the utterance, judges tended to identify several syllables as 'tonic'. In utterances that had a clear 'given/new' structure, either because of clefting or because the information focus was on a (nonfinal) con-

trusted element contradicting a previous element, judgements tended to be more uniform, and when such utterances are presented together with their context in written form, near-uniformity results. (The term 'tonic' as used by Currie is equivalent to our term 'nucleus', and is based on Halliday [1963, 1970], whose terms for prenuclear and nuclear accents are 'pre-tonic' and 'tonic' respectively.) It is difficult to see what theoretical conclusions can be drawn from experiments like these. Some of the conclusions that Currie (1981) in fact draws in a final discussion of the results may well in themselves represent true propositions, but it is arguable whether they can be validly based on her results. In order to accommodate the various 'tonics' that are identified in utterances with a neutral, or perhaps ambiguous 'given/new' structure (the new element occurring as the rightmost lexical item), Currie proposes that a tone group may consist of more than one tone unit, each of which may have a stressed syllable. It is clear that a tone group may contain more than one assigned accent, but while I am not sure if the method applied by Currie would reveal even *THESE* accents (Gussenhoven 1982), it is unlikely to lead to a better understanding of what the nucleus is. It is not unreasonable to see her results as reflecting the unfortunate confusion with which the concept of the tonic is surrounded, in which case they would tell us no more than what we already knew. Since the nucleus is commonly seen — or even described — as 'the most prominent syllable in the tone group', it is not surprising to find that what is essentially a phonological feature is interpreted as an aggregate of (largely) phonetic features (cf. Ladd 1982). It would not be difficult to present listeners with well-formed utterances in which the nucleus is considerably less prominent than one or more prenuclear accents, and so elicit an overwhelming majority of 'tonic' judgements on the prenuclear accent(s). What this illustrates is that — at least for the time span within which judges are required to give their responses — this task overtaxes the introspective powers of the judges to a considerable extent.

The pointlessness of asking linguistically naive or even linguistically non-naive listeners to determine within the time span of a second or two what the nucleus is in an aurally presented utterance is clearly brought out by the results of an experiment with ten native speakers of Dutch. These subjects, who were second-year university students of English, could be assumed to be familiar with the concept of the nucleus from their first-year grammar and phonetics classes. They were given the following two tasks.

1. The Nucleus Test (NT). The subjects were presented with seven utterances which had been produced and recorded on tape by the author. Table 1 lists these utterances, and Figure 1 gives their transcriptions and

Table 1. *Nucleus judgements as a function of F0 height, and the effect of non-nuclear retraction in adverbs substituted in nuclear and non-nuclear positions*

	.91	.09			9 1
1.	<i>Nooit</i> zal hij TOEgeven Never will he give in (always)				altijd
	.21	.50	.14	.07 .07	10
2.	De kapitein is echt <i>op heel veel plaatsen</i> geWEEST The captain is really in very many places been (everywhere)				overall
	.10	.91	.09		10
3.	Dat WEET ik niet. Hij at er in elk geval <i>NOOIT</i> That know I not. He ate there in any case never (always)				altijd
	1.0				10
4.	Problemen waren er <i>NOOIT</i> Problems were there never (everywhere)				overall
	.91	.09			9 1
5.	Ik heb er <i>dikwijls</i> MOEITE mee gehad I have there often trouble with had (always)				altijd
	.83	.17			
6.	Dat heb je <i>VAAK</i> in de uitverkoop That get you often in the sales (occasionally)				(see text)
	.09	.91			8 2
7.	Ga je <i>geregeld</i> naar een DOKter? Go you regularly to a doctor? (occasionally)				wel eens

F0 contours. The utterances were selected with the aim of eliciting a large number of incorrect nucleus markings. To this end, five of them had a wide-range prenuclear accent, followed by a low-range nuclear fall (nos. 1, 2, 3, 4, 5). (Note that the accent on *nooit* in no. 4 is virtually embedded in the slope coming from the prenuclear *-blem-*; note also that without the accent on *nooit* the contour would have steeply shot down to the baseline around *wa-*.) No. 6 had a divided fall-rise, with the fall (marking the nuclear syllable) on a less 'important' word (*vaak* 'often'), and the rise (marking the last syllable of the tail) on an 'important' word (*uitverkoop* 'sales', which word has the word accent on the first syllable [!]). No. 7 had a fairly strong prenuclear accent, followed by a nuclear accent marked by a half-completed fall — also called a 'suspended' fall (Crystal 1969). (A similar incomplete fall occurs in no. 3. Observe that this fall from high to mid is partially masked by the voiceless /t/ of *nooit*. This pattern should

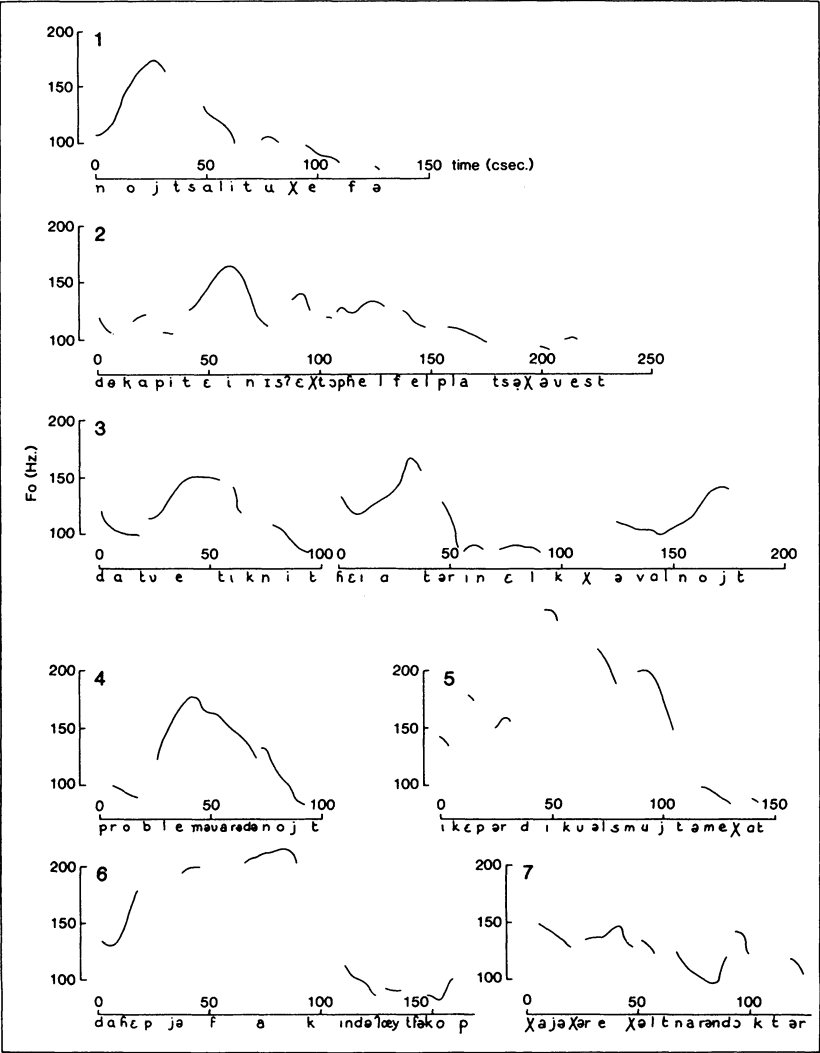


Figure 1. Contours and transcriptions of the utterances in Table 1.

not be confused with the relaxation reversal commonly seen at the ends of rises and falls on fully voiced segments.) Before the NT was given, a number of obvious examples of nucleus placements were discussed, to see if all subjects realized what the test was about. The order of the stimuli in the test was different from that in Table 1, and three further utterances were included to serve as distractors.

2. The Intonation Imitation Task (IIT). In this task the same ten subjects were presented with the ten utterances used in the NT. This time their task was to imitate these utterances as faithfully as they could, but not before they had replaced a particular word in each utterance with a word taken from the set of words that is subject to non-nuclear retraction. On their test sheets the word to be replaced was italicized in the sentence concerned, and the word they had to substitute for it was given in brackets (cf. Table 1). The same three distractors were used. The prediction was that subjects would apply stress shift to the inserted word if that word did not carry the nucleus, i.e. in nos. 1, 2, 5, and 7. Their imitations were scored by three non-naive judges, who were asked to determine which syllable of the seven relevant words was the most accented in that word. (In no. 7 they were asked to mark the most accented syllable of the utterance, see below.) The judges were entirely unanimous, if we ignore the fact that, between them, two of them marked a prefinal syllable as accented IN ADDITION to the final one in three cases.

The results are given in Table 1. The nucleus judgements are given as proportions of the total number of judgements given in the utterance (no. 3 was counted as two utterances). The real nucleus has been capitalized. The scores for the IIT are given in the last column, with the number of times the word accent came on either the first or last syllable given over that syllable.

Both predictions were borne out. In the NT, judges were overwhelmingly misled by the wide-range prenuclear accents and designated them as the nucleus. Yet, from their behavior in the IIT, it is equally clear that they knew — at a tacit level — that another sentence accent came after it: they applied non-nuclear retraction (see in particular nos. 1, 2, and 5). An exception is no. 6. Here, the item *wel eens* had to be slotted into nuclear position. As was observed in section 2.2.3, this adverb cannot have the nucleus, and the task predictably caused havoc in the subjects' responses. Only three subjects responded with the nucleus on *heb* (the only well-formed response), three put it on *wel*, and four on *eens*, which latter pronunciation would of course have been the correct pronunciation if *wel eens* had not been an exceptional item. It may be observed, incidentally, that the agreement between stimulus and imitation with respect to nucleus location is very high (cf. Quirk 1968).

5.0. Summary

In summary, it can be said that the concept of the nucleus is not simply a descriptive expediency that has enabled analysts to break down intona-

tion contours into manageable chunks, to be laid aside when other contour chunks turn out to produce a more elegant analysis, but a phonological reality that is relevant outside the area of pitch contours proper. In order to find evidence for such relevance, the structural descriptions of five rhythm rules were investigated. The English rhythm rule discussed by Liberman and Prince (1977) was found not to be sensitive to the distinction nuclear–non-nuclear, in spite of their claim that it is. Of four Dutch rhythm rules investigated, one turned out to be crucially dependent on whether the fulcrum is nuclear, and its structural description therefore cannot be given without reference to it. It is emphasized that this finding does not support the notion of ‘head contours’ as a separate paradigm, but only that phonological rules may have to ‘know’ whether a sentence accent is final.

No evidence was found in Dutch for the existence of a ‘simple’ fall-rise in addition to a ‘compound’ one, and it was suggested that this distinction has no foundation in English either. It was also suggested that standard descriptions of British English should be amended to include the possibility of associating the rise element of the fall-rise tone with the last syllable of the tone group.

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Instituut Engels-Amerikaans

Katholieke Universiteit

Erasmusplein 1

6500 HD Nijmegen

The Netherlands

Notes

1. I should like to thank Ton Broeders, Gill Brown, Heinz Giegerich, Bob Ladd, Henry Thompson, and an anonymous reviewer for *Linguistics* for their comments on an earlier and somewhat different draft of this article. The contours reported here were produced with the help of standard ILS programs or a program based on Gold and Rabiner (1969) written by Steve Hiller (Figure 1). I thank the staff of the phonetics laboratories of Edinburgh University and Nijmegen University for their help.
2. It should be observed that the research reported in Brown et al. (1980) was done on Edinburgh Scottish English, which dialect of English does not appear to display the rich variety of intonational patterns that standard descriptions of Educated British intonation claim exists in the latter dialect.
3. There is considerable terminological variation among different authors. In addition to the survey in Ladd (1980: 16, 208–209), note Brazil's (1978) term *tonic* for ‘head plus nuclear syllable’, preceding and following stretches being called ‘proclitic’ and ‘enclitic’ respectively. The analytic frameworks show considerable similarity, however.

4. It should be stressed that O'Connor and Arnold's practice of taking combinations of head contours and nuclear tones as their basis for a *semantic* treatment of British English intonation should be seen to reflect the difficulty of assigning meaning to individual tones, head or nuclear, and does not imply that they consider the resulting contours *phonologically* holistic. Their description, inspired by Palmer (1933), has a pedagogical orientation, which, in general, may make it desirable to present larger linguistic structures as learnable units, despite their being made up of elements from different linguistic paradigms.
5. The term 'fulcrum' was suggested to me by Bob Ladd.
6. This is not in fact a restriction Liberman and Prince make. We assume it here for the purpose of discussion, in order to avoid running into problems that are unrelated to either stress shift or the nucleus.
7. The difference between bimorphemic and monomorphemic *onvermijdelijk* is perhaps better brought out by the example that the Anglist R. C. J. Born once used to illustrate the difference between *unavoidable* and *inevitable*: *Hij was onvermijdelijk te laat* ('He was unavoidably late' and 'He was inevitably late'). Written evidence for rhetorical retraction appeared in *Onze Taal* (1981) 50 (12): 129, where it said 'Een spontaan applausje voor deze dichters!' ('A spontaneous [little] applause for these poets!') in the column 't *Rijmschap*. *Spontaan* is monomorphemic and has the word accent on the second syllable.
8. The case of *wel eens* was brought to my attention by Ton Broeders.
9. I here ignore the interesting complication that the more literal — and longer — translation *Mijn moeder is in Sheffield geboren* has a different word order from the English, and that this word order is greatly preferred to ... *is geboren in Sheffield* if this stretch is unaccented, but that the latter word order is quite natural if *Sheffield* is accented. This fact in itself is evidence that in Dutch *Sheffield* is non-nuclear in either meaning of (34).
10. Lee (1956) comes close to making this distinction. After cataloguing the widely different interpretations of fall-rise patterns in the (British English) literature, he proposes a distinction between a fall-rise TONE and a fall-rise SEQUENCE. In the tone, so he says, the rise element generally begins immediately after the fall, while in the sequence the rise occurs on a later word, with the intervening syllables generally remaining low in pitch, or rising just a little. The semantic distinctions noted between the two patterns are either similar to that claimed by O'Connor and Arnold for the examples here given as (32) and (33) (e.g. *She doesn't speak to anybody she meets along the road*, i.e. 'only to certain people', versus *She doesn't speak to anybody she meets along the, road*, i.e. 'she is generally silent when traveling'), or would seem to refer to differences between fall-rise nuclei and tone groups having a falling head followed by a rising nuclear tone. This latter point was also made by Schubiger (1956). Note that in neither the tone nor the sequence is the rise associated by Lee with the last SYLLABLE. Schubiger also (1956) says that in the sequence, the rise 'often quite slight (comes) on the weightiest word (in the tail)'. The descriptive bias toward pattern (a) may have a historical explanation. Often, tone groups with a fall-rise nuclear tone followed by a tail that is separated from the preceding material by a comma in writing are characterized as sequences of tone groups (cf. most of the examples in Palmer [1922: 91–92] and at least some in Armstrong and Ward [1931: 35]). Evidently, in such an analysis, the second tone would have to be associated with a word rather than a syllable. In Kingdon, the two tones have become a 'divided Tone III', a fall-rise spread over two words: 'The rising part (comes on a later word) which is included in the feeling conveyed by Tone III, but to which only partial emphasis need be given' (1958:

20). In addition, it could be the case that pattern (a) should be interpreted dialectally, such that it was either more common earlier in this century, or is more strictly part of a narrowly defined RP. Jones (1962: 303) even says, 'When there is no stressed or semi-stressed syllable after the emphatic word the terminal rise begins at the syllable following the emphatic word', which description squares with the representation given in Palmer (1922: 10).

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